

DATA ENGINEERING COURSE

Employee Dataset Cleaning Challenge

Practicing Missing Value Handling, Deduplication, Renaming, Type Conversion and String Standardization with Pandas

ASSIGNMENT

INSTRUCTOR

Tanishq Tyagi

Before You Begin

- Read every question carefully before writing your code.
- Use the pandas library to complete all tasks in this assignment.
- Write clean, readable code with meaningful variable names and proper indentation.
- Complete the questions in the order given, since later steps build on the results of earlier steps.
- Print or display the output at each step so your work can be verified.
- Submit your solution as a single, well-commented Python script or notebook.

Dataset

Use the following employee dataset for all questions in this assignment.

STARTING DATASET

```
employees = {
    "Employee Name ": ["amit", "NEHA ", "Rohan", None, "amit", "Simran"],
    "Age": [25, np.nan, 30, 28, 25, np.nan],
    "Salary": [50000, 65000, np.nan, 45000, 50000, 80000],
    "Department": [" it", "HR ", "Finance", "IT", " it", "Hr"],
    "City": ["Delhi ", " mumbai", "DELHI", "Pune", "Delhi ", " Bangalore"],
}

df = pd.DataFrame(employees)
print(df)
```

Employee Dataset Cleaning Challenge

Section A: Identifying and Handling Missing Values

- 1 Identify all missing values in the dataset.
- 2 Count the missing values in each column.
- 3 Replace the missing values in the Age column using the mean of the column.
- 4 Replace the missing values in the Salary column using the median of the column.

Section B: Removing Duplicates

- 5 Remove duplicate rows from the dataset.

Section C: Renaming and Standardizing Columns

- 6 Rename the column " Employee Name " to "employee_name".
- 7 Standardize all column names by removing extra spaces, converting them to lowercase, and replacing spaces with underscores.

Section D: Standardizing Text Columns

- 8 Standardize the employee_name column by removing extra spaces and converting values to Title Case.
- 9 Standardize the department column by removing extra spaces and converting values to Uppercase.
- 10 Standardize the city column by removing extra spaces and converting values to Title Case.

Section E: Data Types and Feature Creation

11 Convert the Age column to integer type.

12 Create a new column called salary_category, where a salary greater than or equal to 60,000 is labeled "High" and a salary less than 60,000 is labeled "Low".

Section F: Final Output

13 Display the final cleaned dataset.

14 Display the data types of each column in the final cleaned dataset.
